



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23			IV / IV - B.Tech. I - Semester						
ELECTRONICS & COMMUNICATION ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23EC4101	Cellular and Mobile Communications	PC	3	0	0	3	30	70	100
B23HS4104	Management Science	HS	2	0	0	2	30	70	100
#PE-IV	Professional Elective -IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective -V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23EC4113	Digital Signal and Image Processing Lab	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	-	30	-	30
B23EC4114	Evaluation of Industry Internship /Mini Project	PR	-	-	-	2	-	50	50
TOTAL			19	01	02	21	240	540	780

	Course Code	Course
#PE-IV	B23EC4102	Analog VLSI Layout Techniques
	B23EC4103	Embedded Communication Protocols & Interfaces
	B23EC4104	Coding Theory & Applications
	B23EC4105	Optical Fiber Communication
	B23EC4106	DSP Processors and Architectures
	B23EC4107	Design for Testability
#PE-V	B23EC4108	Physical Design
	B23EC4109	Internet of Things & its Applications
	B23EC4110	Digital Image Processing
	B23EC4111	Satellite Communication
	B23EC4112	Low Power VLSI Design
#OE-III	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or EEE or IT or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or EEE or IT or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4101	PC	3	--	--	3	30	70	3 Hrs.
CELLULAR & MOBILE COMMUNICATIONS								
(for ECE)								
Course Objectives:Students are expected to								
1.	Understand the fundamentals of cellular communication systems, including limitations of conventional systems, types of interferences, and handoff mechanisms.							
2.	Apply concepts of frequency management, channel assignment, and multiple access techniques in mobile communication systems.							
3.	Develop skills to explore, analyze, and present emerging trends and technologies in mobile and cellular communications.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the limitations of conventional mobile telephone systems and the concepts of cellular systems.							K2
2.	Describe the different types of interferences influencing cellular and mobile communications							K2
3.	Illustrate frequency management, channel assignment, and handoff mechanisms in cellular systems							K2
4.	Analyze the concept and performance of handoff in cellular systems							K3
5.	Evaluate multiple access schemes and present insights on emerging mobile and cellular technologies.							K3
SYLLABUS								
UNIT-I (12Hrs)	CELLULAR SYSTEMS: Limitations of Conventional System, Basic Cellular Mobile System, First, second, third and fourth Generation cellular wireless systems. Operation of Cellular System. Uniqueness of Mobile Radio Environment –Fading, coherence bandwidth, Doppler Spread. FUNDAMENTALS OF CELLULAR RADIO SYSTEM DESIGN: concept of frequency reuse channels, Co-channel Interference, Co-channel Interference Reduction Factor, desired C/I from a normal case in a Omni directional Antenna system, Trunking and grade of service.							
UNIT-II (10Hrs)	CO-CHANNEL& NON- CO-CHANNEL INTERFERENCE: Measurement of Real Time Co-Channel Interference, Non co-channel interference-adjacent channel interference, Near End far end interference, effect on coverage and interference by power decrease, antenna height decrease.							
UNIT-III (10Hrs)	FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT: Numbering and grouping, setup access and paging channels, channel assignments to cell sites and mobile							

	units, channel sharing and borrowing, sectorization, overlaid cells, non-fixed channel assignment. CELL COVERAGE FOR SIGNAL AND TRAFFIC : Signal reflections in flat and hilly terrain, effect of human made structures, phase difference between direct and reflected paths, straightline path loss slope, general formula for mobile propagation over water and flat open area, near and long-distance propagation.
UNIT-IV (10 Hrs)	CELLSITE AND MOBILE ANTENNAS: Sum and difference patterns, omni directional antennas, directional antennas for interference reduction, space diversity antennas, umbrella pattern antennas, minimum separation of cell site antennas, highgain antennas. HANDOFFS: Handoff Initiation, types of handoff, delaying handoff, advantages of Handoff, power difference hand off, forced hand off, mobile assisted and soft and Hard handoffs. Intersystem and Intra system handoffs, dropped call rates and their evaluation.
UNIT-V (8Hrs)	DIGITAL CELLULAR NETWORKS AND MULTIPLE ACCESS SCHEME: GSM architecture, GSM channels, TDMA, FDMA, and CDMA. Introduction to MIMO systems Principles of CDMA cellular systems, Introduction of 5G & 6G.
Textbooks:	
1.	Mobile Cellular Telecommunications – W.C.Y. Lee, Tata McGraw Hill, 2 nd Edn., 2006.
2.	T.S.Rappaport, “Wireless Communications–Principles and Practice”(2 nd edition) Pearson, 2010, ISBN 9788131731864.
Reference Books:	
1.	Principles of Mobile Communications–Gordon L. Stuber, Springer International 2 nd Edition, 2001
2.	Modern Wireless Communication–Simon Haykin Michael Moher, Persons Education,2005.
3.	Wireless Communication theory and Techniques, Asrar U. H. Sheikh Springer,2004
e-Resources	
1.	https://www.youtube.com/watch?v=f2wlHL1Sok8&list=PLuv3GM6gsE3ypUYh43pPuZsXxJV_G1e7F
2.	https://archive.nptel.ac.in/courses/128/108/128108016/

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B23HS4104	HS	3	--	--	3	30	70	3 Hrs.
MANAGEMENT SCIENCE								
(For ECE)								
Course Objectives:								
1.	To impart knowledge of source and function of values.							
2.	Demonstrate an understanding of the importance of values, ethics, and social responsibility for the self and for contemporary society.							
3.	Reflect on how values shape personal and community ethics and decision-making. An ability to function effectively on multi-disciplinary teams (team work).							
4.	Inculcate professional, ethical, legal, security and social issues and responsibilities.							
5	Gain personal and professional insight into organizational behaviour, diversity, personalities, goal setting, motivation, empowerment, and leadership style.							
Course Outcomes: By the end of the course, the student should have the ability to:								
S.No	Outcome							Knowledge Level
1.	Summarize management and motivation theories to renovate the practice of management							K2
2.	Concepts of plant location, plant layout, work study , work measurement ,various types of charts, differences in the time study							K2
3.	Compute Optimum to apply project management techniques (PERT/CPM) and quality control methods to plan, control, and improve organizational performance."							K3
4.	Apply principles of materials management to effectively control inventory, manage stores, and optimize purchasing for organizational efficiency							K4
5.	Apply HRM functions and job evaluation methods to establish fair compensation practices.							K4
SYLLABUS								
UNIT-I (10 Hrs)	CONCEPTS OF MANAGEMENT AND ORGANISATION: Functions of management, evolution of management thought, Taylor's scientific management, fayol's principles of management, Hertzberg's Maslow's hierarchy of human needs, theory x and y, Hawthorne experiment, morale, motivation, working environmental conditions, systems approach to management. DESIGNING ORGANISATIONAL STRUCTURES: Basic concepts related to organization- types of organization structures.							
UNIT-II (10 Hrs)	PLANTLOCATION: Definition, factors affecting the plant location, comparison of rural and urban sites. Plant layout - definition, objectives, types of plant layout, various data analyzing forms travel chart.							

	WORKSTUDY: Definition, objectives, method study-definition, objectives, steps involved-various types of associated charts, Work measurement- definition, time study, steps involved, Work Sampling -definition, steps involved, standard time calculations.
UNIT-III (10 Hrs)	INTRODUCTION TO PERT /CPM: Project management, network modeling- probabilistic model, various types of activity times estimation, programme evaluation review techniques, critical path, probability of completing the project, deterministic model, critical path method (CPM),critical path calculation, crashing of simple of networks. INSPECTION AND QUALITY CONTROL: Types of inspections, statistical quality control, techniques, variables and attributes, variable control charts, and Rcharts, pcharts and ccharts.
UNIT-IV (8 Hrs)	MATERIALS MANAGEMENT: Objectives, inventory functions, types, associated costs, inventory classification techniques-ABC and VED analysis. Inventory control systems, continuous review system, periodical review system. Stores management and stores records. Purchase management, duties of purchase of manager.
UNIT-V (12 Hrs)	INTRODUCTION TO HUMAN RESOURCE MANAGEMENT: Functions of HRM, job evaluation, different types of evaluation methods. Job description, merit rating, different methods of merit ratings, wage incentives, different types of wage incentives schemes. Marketing, selling, marketing mix, product lifecycle.
Text Books:	
1.	P.Khanna(2004),Industrial Engineering andManagement, DhanpatRai, New Delhi.
2.	Management Science by Aryasri: Publisher: Tata McGraw Hill 2009
Reference Books:	
1.	Stoner, Freeman (2005), Gilbert, Management, 6th edition, Pearson Education, New Delhi.
2.	PannerSelvam (2004), Production and Operations Management, Prentice Hall of India, New Delhi.
3.	Ralph M. Barnes (2004), Motion and Time Studies, John Wiley and Sons.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4102	PE	3	--	--	3	30	70	3 Hrs.
ANALOG VLSI LAYOUT TECHNIQUES								
(For ECE)								
Course Objectives:Students are expected to								
1.	Impart knowledge of Analog IC layout principles and layout-related reliability issues in the context of both planar MOSFET and FinFET technologies.							
2.	Develop practical skills in CAD-based Analog layout design of precision circuits.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Understand IC technology, MOS transistor characteristics, fabrication process and scaling of MOS circuits.							K2
2	Understand layout principles and optimize device configurations to enhance Analog circuit accuracy and noise performance.							K2
3	Apply key reliability challenges and appropriate layout techniques to improve circuit robustness and performance in analog IC layout.							K3
4	Interpret various advanced analog layout techniques like device isolation and multi-domain power structures to enhance circuit performance and reliability.							K2
5	Apply FinFET technologies, well structures, fabrication challenges and analog layout techniques including common centroid and interdigitation.							K3
SYLLABUS								
UNIT-I (12 Hrs)	Introduction to IC Technology and CMOS Fabrication: NMOS, PMOS and CMOS. MOSFET Characteristics: Ids–Vds relationships, threshold voltage, and transconductance. CMOS Circuits: Inverter operation and latch-up in CMOS. VLSI Layout Basics: MOS layers, stick diagrams, and design rules (lambda-based rules). Scaling of MOS circuits: scaling models and scaling factors.							
UNIT-II (09 Hrs)	Analog vs Digital, noise in MOSFET, Guard ring, Devices sizes in Analog Layout, Fingers and Multipliers. Analog Layout: Matching, Methods of Matching, Inter-digitized, Common Centroid, Guidelines for Matching.							
UNIT-III (08 Hrs)	Analog Layout Challenges: Electromigration, Electromigration - Flow, Electromigration – Layout care, IR Voltage drop, Latch-Up, ESD, Antenna Effect, Coupling & Shielding.							
UNIT-IV (08 Hrs)	Shallow Trench Isolation, Well Proximity Effect, Multi Domain Power and Ground, Cross Section of cascode devices, Deep Nwell, Separate potential Nwell.							

UNIT-V (10 Hrs)	Planar MOSFET Vs FinFET ,FinFET technology, Fabrication and design challenges in FinFET technology, well structures. Opamp layout with common centroid matching, Current mirror with Interdigitation Matching, Differential pair circuit.
Textbooks:	
1.	CMOS Analog Circuit Designby Phillip E. Allen and Douglas R. Holberg
2.	"The Art of Analog Layout" by Alan Hastings
Reference Books:	
1.	"CMOS VLSI Design: A Circuits and Systems Perspective"by Neil H.E. Weste, David Harris
2.	"Design of Analog CMOS Integrated Circuits"by Behzad Razavi
3.	"FinFETs and Other Multi-Gate Transistors"by J.-P. Colinge
e-Resources	
1.	https://www.scribd.com/document/674230335/Analog-Layout-Design-Book
2.	https://www.yumpu.com/en/document/view/27132420/analog-layout-techniques



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4103	PE	3	--	--	3	30	70	3 Hrs.
EMBEDDED COMMUNICATION PROTOCOLS & INTERFACES								
(For ECE)								
Course Objectives:Students are expected to								
1.	Understand the Embedded Communication Protocols							
2.	Develop a network based on Embedded systems							
3.	Know about interfaces and buses							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrates the basic embedded communication protocols							K2
2.	Illustrates the Serial & Parallel communication protocols							K2
3.	Explain the network protocols							K2
4.	Illustrates the networked embedded systems							K2
5.	Explain the Wireless Communication protocols							K2
SYLLABUS								
UNIT-I (8Hrs)	Basic Embedded Communication Protocols: I/O types & examples, serial communication devices-synchronous and asynchronous communications, RS232C RS485 communication, UART, HDLC protocol, Serial data communication using SPI, SCI & SI Ports, I2C protocol, parallel port interfacing examples.							
UNIT-II (8 Hrs)	Serial & Parallel communication protocols: serial bus communication protocols- I2C Bus, CAN Bus, USB Bus, Fire Wire, Parallel Bus Device protocols-ISA Bus, PCI & PCI/X Buses, ARM Bus, Timer & Counting Devices, Watchdog timer, Realtime clock.							
UNIT-III (8Hrs)	Internet Enabled Systems-Network Protocols: Hyper-Text Transfer Protocol (HTTP), Transport Control Protocol (TCP), Internet Protocol (IP), Ethernet-Network elements, Bus and Star Topologies, protocol CSMA/CD&CA, IEEE 802.3 and its different versions, Ethernet Addressing, Cables, Connections, Hubs, switches, and Bridges.							
UNIT-IV (8 Hrs)	Networked Embedded systems: An Overview, Networking of Embedded Systems Automotive Networked Embedded Systems, Networks Embedded Systems in Industrial Automation, Networked Embedded Systems in Building Automation, Introduction to Wireless Sensor Networks-Typical Scenarios and Applications, Design Challenges.							
UNIT-V (8 Hrs)	Wireless Communication Protocols: Fundamentals of (wireless) MAC protocols, Fixed assignment MAC protocols (TDMA, FDMA, CDMA & SDMA), Infrared Data							

	Association (IrDA), Bluetooth, IEEE 802.11 standard, ZIGBEE protocol.
Textbooks:	
1.	Embedded Systems Architecture, Programming and Design- <u>Raj Kamal</u> , <u>Tata McGraw-Hill</u>
2.	Networked Embedded Systems-Richard Zurawski, CRC Press
Reference Books:	
1.	Embedded Systems Design: A Unified Hardware/Software Introduction - Frank Vahid, Tony Givargis, John & Wiley Publications, 2002
2.	Wireless Communications & Networking- Vijay K. Garg, Morgan Kaufmann Publishers
e-Resources	
1.	https://archive.nptel.ac.in/courses/108/102/108102169/
2.	https://archive.nptel.ac.in/courses/108/102/108102045/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4104	PE	3	--	--	3	30	70	3 Hrs.
CODING THEORY AND APPLICATIONS								
(For ECE)								
Course Objectives:Students are expected								
1.	To get exposed to information and entropy and to Learn measurement of information and errors							
2.	To obtain knowledge in designing various source codes and channel codes							
3.	To design encoders and decoders for block and cyclic codes							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply entropy and source coding techniques like Shannon-Fano and Huffman coding to measure and compress information.							K3
2.	Analyze discrete communication channels using channel matrices, mutual information, and channel capacity calculations.							K4
3.	Implement linear block coding techniques for error detection and correction, including Hamming code.							K3
4.	Design cyclic code generators and evaluate their error-correcting capabilities using generator polynomials and syndrome decoding.							K4
5.	Design and analyze convolutional encoders using time and transform domain approaches and evaluate their error correction performance using the Viterbi algorithm in comparison with block codes.							K4
SYLLABUS								
UNIT-I (08 Hrs)	Information Theory: Introduction, Measure of information, Information content of message, Entropy, Information rate, Joint and conditional entropies, Source coding theorem, Shannon-fano coding, Huffman coding.							
UNIT-II (08 Hrs)	Information Channels: Communication Channels, Discrete Communication channels Channel Matrix, Joint probability Matrix, System Entropies. Mutual Information, Channel Capacity, Discrete memory less channels – BSC, Shannon limit Channel Capacity of Binary Symmetric Channel.							
UNIT -III (08 Hrs)	Error Control Coding: Introduction to Linear Block Codes, Syndrome and Error Detection, Minimum Distance of a Block code, Error-Detecting and Error-correcting Capabilities of a Block code, methods of Controlling Errors, Single error correction Hamming code.							
UNIT-IV	Binary Cyclic Codes: Algebraic Structure of Cyclic Codes, generation of cyclic codes,							

(08 Hrs)	Encoding using an (n-k) Bit Shift register, Syndrome Calculation, generation of cyclic codes using generator methods, Introduction to BCH codes, RS codes and GOLAY codes, ARQ's.
UNIT-V (08 Hrs)	Convolution Codes: Convolution Encoder, Time domain approach, Transform domain approach, Code Tree, Trellis and State Diagram, The Viterbi Algorithm, difference between convolution codes and block codes, burst error corrections, turbo encoder, introduction to concatenated codes.
Textbooks:	
1.	Digital and Analog Communication Systems, K. Sam Shanmugam, John Wiley India Pvt Ltd, 1996
2.	Digital Communication, Simon Haykin, John Wiley India Pvt Ltd, 2008
Reference Books:	
1.	ITC and Cryptography Ranjan Bose, TMH, II edition, 2007
2.	Digital Communications- Fundamentals and Applications Bernard Sklar, Second Edition, Pearson Education, 2016, ISBN: 9780134724058.
e-Resources	
1.	https://nptel.ac.in/courses/108102117



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4105	PE	3	--	--	3	30	70	3 Hrs.
OPTICAL FIBER COMMUNICATION								
(For ECE)								
Course Objectives:Students are expected to								
1.	Learn the basic elements of optical fiber transmission link, fiber modes, configurations and structures.							
2.	Understand the different kind of losses, signal distortion, Modes of fibers							
3.	Learn the various optical sources, materials and fiber splicing.							
4.	Learn the fiber optical receivers and noise performance in photo detector.							
5.	Learn link budget, WDM, solitons and SONET/SDH network.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Demonstrate an understanding of optical fiber communication link, structure, propagation and transmission properties of an optical fiber.							K3
2.	Design and build optical fiber experiments in the laboratory, and learn how to calculate electromagnetic modes in waveguides, the amount of light lost going through an optical system, dispersion of optical fibers.							K3
3.	Choose the optical cables for better communication with minimum losses							K3
4.	Use different types of photo detectors and optical test equipment to analyze optical fiber and light wave systems.							K3
5.	Design, build, and demonstrate optical fiber experiments in the laboratory.							K3
SYLLABUS								
UNIT-I (10Hrs)	Overview of optical fiber communication –Historical development, advantages of optical fiber communications, Introduction to Raytheory transmission, Total Internal Reflection, Acceptance angle, Numerical Aperture, Skew rays, Cylindrical fibers- Modes, V-number, Mode coupling, Step Index fiber, Graded Index fiber, Single mode fiber-Cutoff wave length, Related problems.							
UNIT-II (10 Hrs)	Fiber materials - Glass, Halide, Active glass, Plastic optical fibers. Signal distortion in optical fibers-Attenuation, Absorption, Scattering and Bending losses, Group delay, Types of Dispersion:- Material dispersion, Wave-guide dispersion, Intermodal dispersion, Pulse broadening in Graded index fiber, Related problems.							
UNIT-III (10 Hrs)	Optical fiber Connectors- Connector types, Single modefiber connectors, Connect or returnloss, Fiber Splicing-Splicing techniques, Splicing single mode fibers, Fiber alignment and joint loss, single mode fiber joints.							

UNIT-IV (10 Hrs)	Optical sources -LEDs, Structures, Quantum efficiency, Power, Modulation, Power bandwidth product. Injection Laser Diodes- Modes, Threshold conditions, External quantum efficiency, Laser diode rate equations, Reliability of LED&ILD, Optical detectors- Physical principles of PIN and APD, Detector response time, Comparison of Photo detectors.
UNIT-V (10 Hrs)	Source to fiber power launching - Output patterns, Power coupling, Power launching, Equilibrium Numerical Aperture, Laser diode to fiber coupling, Optical receiver, Optical receiver operation, Receiver configuration, Digital receiver performance, Probability of Error, Quantum limit, Analog receivers. Optical system design - Point-to- point links- Component choice and considerations, Link power budget, Rise time budget with examples, WDM, Necessity, Principles, Eye pattern.
Textbooks:	
1.	Fiber Optic Communications Technology – D.K. Mynbaev and Lowell L. Scheiner, Pearson Education, 2009.
2.	Optical Fiber Communications – Gerd Keiser, Mc Graw-Hill International edition, 4th Edition, 2000.
Reference Books:	
1.	Fiber Optic Communications – Joseph C. Palais, 4th Edition, Pearson Education, 2004.
e-Resources	
1.	https://www.thefoa.org/tech/ref/basic/fiber.html
2.	https://media.wiley.com/product_ancillary/17/04705051/DOWNLOAD/Fiber_Optic_Communication_Systems.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4106	PE	3	--	--	3	30	70	3 Hrs.
DSP PROCESSORS AND ARCHITECTURES								
(For ECE)								
Course Objectives:Students are expected to								
1.	Understand the fundamental principles of Digital Signal Processing (DSP), including discrete-time signals, sampling, and frequency analysis using DFT and FFT.							
2.	Analyze the architecture and functioning of programmable DSP processors, including memory organization, addressing modes, and instruction execution.							
3.	Develop the ability to implement DSP systems by handling computational accuracy issues interfacing memory and I/O peripherals with DSP devices.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the basics of Digital Signal Processing and transforms.							K2
2.	Able to distinguish between the architectural features of general-purpose processors andDSP processors.							K3
3.	Understand the architectures of TMS320C54xx devices and ADSP 2100 DSP devices.							K3
4.	Able to write simple assembly language programs using instruction set of TMS320C54xx.							K3
5.	Can interface various devices to DSP Processors.							K2
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Digital Signal Processing: Introduction, A Digital signal-processing system, The sampling process, Discrete time sequences, Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT), Linear time-invariant systems, Digital filters, Decimation and interpolation. Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.							
UNIT-II (10 Hrs)	Architectures for Programmable DSP Devices: Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation Unit, Programmability and Program Execution, Speed Issues, Features for External interfacing.							
UNIT-III (10 Hrs)	Programmable Digital Signal Processors: Commercial digital signal processing devices, Data Addressing modes of TMS320C54XX DSPs, data Addressing modes of TMS320C54XXProcessors, Memory space of TMS320C54XX processors, program							

	control, TMS320C54XX instructions and programming, On-Chip Peripherals, Interrupts of TMS320C54XX processors, pipeline Operation of TMS320C54XX Processors.
UNIT-IV (10 Hrs)	Analog Devices Family of DSP Devices: Analog Devices Family of DSP Devices ALU and MAC block diagram, Shifter Instruction, Base Architecture of ADSP 2100, ADSP2181 high performance processor. Introduction to Blackfin Processor- The Blackfin Processor, Introduction to Micro signal Architecture, Overview of Hardware Processing Units and Register files, Address Arithmetic Unit, Control Unit, Bus Architecture and Memory, Basic Peripherals.
UNIT-V (10 Hrs)	Interfacing Memory and I/O Peripherals to Programmable DSP Devices: Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA).
Textbooks:	
1.	Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.
2.	A Practical Approach To Digital Signal Processing - K Padmanabhan, R. Vijayarajeswaran, Ananthi. S, New Age International, 2006/2009.
Reference Books:	
1.	Digital Signal Processors, Architecture, Programming and Applications–B. Venkataramani and M. Bhaskar, 2002, TMH.
2.	DSP Processor Fundamentals: Architectures & Features – Lapsley et al., S. Chand & Co. Digital Signal Processing Applications Using the ADSP-2100 Family, Prentice-Hall, Inc.
e-Resources:	
1.	https://www.dspguide.com/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4107	PE	3	--	--	3	30	70	3 Hrs.
DESIGN FOR TESTABILITY								
(For ECE)								
Course Objectives:Students are expected								
1.	To give exposure on the Testing, Fault modelling.							
2.	To give exposure on Fault simulation and Testability measures.							
3.	To give exposure on Built in Self-Test and Boundary Scan Test Instructions.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Acquire verification knowledge and test evaluation							K2
2.	Design for testability rules and techniques.							K3
3.	Utilize the scan architectures for different digital circuits.							K3
4.	Acquire the knowledge of design of built-in-self test.							K3
5.	Interpret Boundary Scan Standards in Testing							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Testing: Testing Philosophy, Role of Testing, Digital and Analog VLSI Testing, VLSI Technology Trends affecting Testing, Types of Testing, Fault Modeling: Defects, Errors and Faults, Functional Versus Structural Testing, Levels of Fault Models, Single Stuck-at Fault.							
UNIT-II (10 Hrs)	Logic and Fault Simulation: Simulation for Design Verification and Test Evaluation, Modeling Circuits for Simulation, Algorithms for True-value Simulation, Algorithms for Fault Simulation, ATPG.							
UNIT-III (10 Hrs)	Testability Measures: SCOAP Controllability and Observability, High Level Testability Measures, Digital DFT and Scan Design: Ad-Hoc DFT Methods, Scan Design, Partial-Scan Design, Variations of Scan.							
UNIT-IV (10 Hrs)	Built-In Self-Test: The Economic Case for BIST, Random Logic BIST: Definitions, BIST Process, Pattern Generation, Response Compaction, Built-In Logic Block Observers, Test-Per-Clock, Test-Per-Scan BIST Systems, Circular Self Test Path System, Memory BIST, Delay Fault BIST.							
UNIT-V (10 Hrs)	Boundary Scan Standard: Motivation, System Configuration with Boundary Scan: TAP Controller and Port, Boundary Scan Test Instructions, Pin Constraints of the Standard, Boundary Scan Description Language: BDSL Description Components, Pin Descriptions.							

Textbooks:	
1.	M.L. Bushnell, V. D. Agrawal, “Essential of Electronic Testing for Digital, Memory and Mixed Signal VLSI Circuits”, Kluwer Academic Publishers
2.	P. K. Lala, “Digital Circuits Testing and Testability”, Academic Press
Reference Books:	
1.	M. Abramovici, M. A. Breuer and A.D Friedman, Digital Systems and Testable Design”, Jaico Publishing House
e-Resources	
1.	https://books.google.co.in/books?id=bw16LHRVH7IC&printsec=copyright&redir_esc=y#v=onepage&q&f=false
2.	https://books.google.co.in/books?id=UTrtBwAAQBAJ&pg=PR7&source=gbv_selected_pages&cad=3#v=onepage&q&f=false



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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4108	PE	3	--	--	3	30	70	3 Hrs.
PHYSICAL DESIGN								
(For ECE)								
Course Objectives: Students are expected to								
1.	Introduce students to the fundamentals of VLSI physical design and design flow.							
2.	Develop understanding of floor planning, placement, and clock tree synthesis techniques.							
3.	Enable students to analyze and optimize design parameters such as area, power, and timing.							
4.	Provide knowledge of routing, interconnect modeling, and timing-driven design methodologies.							
5.	Cultivate skills in static timing analysis and performing basic signoff checks in physical design.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the fundamentals of VLSI physical design and ASIC design flow.							K2
2.	Apply floor planning techniques, aspect ratio, and power planning concepts in physical design.							K3
3.	Analyze placement strategies, congestion, and clock tree synthesis (CTS) for optimized design.							K3
4.	Analyze routing techniques, interconnect effects, and timing optimization methods.							K3
5.	Describe static timing analysis, timing checks, and signoff verification basics.							K2
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Introduction to VLSI Physical Design, Need for Physical Design, ASIC Design Flow, Physical Design Automation, Design flow stages (Behavioral, Structural, Physical), Inputs to block PD implementation (basic), Design import (basic).							
UNIT-II (10 Hrs)	Floorplanning: Full chip floorplanning and block level floorplanning, Aspect ratio, Die and core utilization, Macro placement guidelines, Power planning concepts: Power consumption in IC, Static power, Dynamic power, Power plan goals, Power delivery and IR drop (basic).							
UNIT-III (10 Hrs)	Placement concepts: Goals of placement, Types of placement, Placement optimization techniques, Congestion analysis, Timing analysis. Clock Tree Synthesis (CTS): CTS specification, CTS QoR, CTS distribution methodologies, Timing-driven placement.							
UNIT-IV (10 Hrs)	Routing concepts: Global routing, Detailed routing, Routing challenges and objectives, Timing-driven routing, Interconnect modeling (basic), Timing closure concepts,							

	Performance-driven design flow, Timing optimization techniques.
UNIT-V (10 Hrs)	Fundamentals of Static Timing Analysis (STA): Timing analysis, Timing checks (Setup), Timing report, Hold checks (Hold), Multiple clock domains (basic), Extraction (basic), Signoff checks overview (Design Rule Check – DRC basics).
Textbooks:	
1.	J. Bhasker, Rakesh Chadha, Static Timing Analysis for Nanometer Designs: A Practical Approach. Springer US, April 3, 2009. ISBN: 9780387938202, 0387938206.
2.	Khosrow Golshan, Physical Design Essentials: An ASIC Design Implementation Perspective, Springer Publications.
Reference Books:	
1.	Michael J. S. Smith, Application-Specific Integrated Circuits, Addison-Wesley, 1997
2.	Fundamentals of Digital Circuits by A.Anand Kumar, PHI learning Pvt. Ltd.
3.	Fundamentals of Logic Design, Charles H Roth, Thomson Publications, 5th edition, 2009
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc26_ee90/preview
2.	https://onlinecourses.nptel.ac.in/noc26_ee70/preview



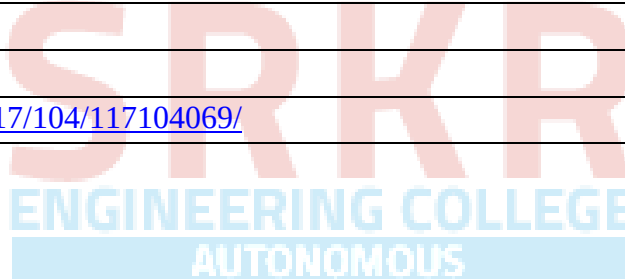
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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4109	PE	3	--	--	3	30	70	3 Hrs.
INTERNET OF THINGS & ITS APPLICATIONS								
(For ECE)								
Course Objectives:Students are expected to								
1.	Understand the fundamental concepts of IoT							
2.	Explore the networking and communication protocols							
3.	Study the design methodology and different IoT hardware platforms.							
4.	Study the basics of IoT Data Analytics and supporting services.							
5.	Study real-world applications and the impact of IoT on Industry 4.0, smart infrastructure, and consumer electronics.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the basics of IoT.							K2
2.	Analyze and select appropriate communication protocols based on network constraints and application requirements.							K4
3.	Understand design methodology and hardware platforms involved in IoT.							K3
4.	Understand how to analyze and organize the data.							K4
5.	Compare IoT Applications in Industrial & real-world.							K4
SYLLABUS								
UNIT-I (10Hrs)	FUNDAMENTALS OF IoT- Evolution of Internet of Things, Enabling Technologies, M2M Communication, IoT World Forum (IoTWF) standardized architecture, Simplified IoT Architecture, Core IoT Functional Stack, Fog, Edge and Cloud in IoT, Functional blocks of an IoT ecosystem, Sensors, Actuators, Smart Objects and Connecting Smart Objects.							
UNIT-II (10 Hrs)	IoT PROTOCOLS- IoT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.11ah and Lora WAN, Network Layer: IP versions, Constrained Nodes and Constrained Networks,6LoWPAN, Application Transport Methods: SCADA, Application Layer Protocols: CoAP and MQTT.							
UNIT-III (10 Hrs)	DESIGN AND DEVELOPMENT- Design Methodology, Embedded computing logic, Microcontroller, System on Chips, IoT system building blocks IoT Platform overview: Overview of IoT supported Hardware platforms such as: Raspberry pi, Arduino Board details.							
UNIT-IV	DATA ANALYTICS AND SUPPORTING SERVICES:							

(10 Hrs)	Data Analytics: Introduction, Structured Versus Unstructured Data, Data in Motion versus Data at Rest, IoT Data Analytics Challenges, Data Acquiring, Organizing in IoT/M2M. Supporting Services: Computing Using a Cloud Platform for IoT/M2M Applications/Services, Everything as a service and Cloud Service Models.
UNIT-V (10 Hrs)	CASE STUDIES/INDUSTRIAL APPLICATIONS: IoT applications in home, infrastructures, buildings, security, Industries, Home appliances, other IoT electronic equipments, Industry 4.0 concepts.
Textbooks:	
1.	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017
2.	Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015
Reference Books:	
1.	From Machine-to-Machine to the Internet of Things – Introduction to a New Age of Intelligence”, Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stamatis, Karnouskos, Stefan Avesand. David Boyle and Elsevier, 2014.
2.	Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Michahelles and Florian (Eds), Springer, 2011.
3.	Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill Higher Education.
e-Resources	
1.	https://www.codemag.com/article/1607071/Introduction-to-IoT

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4110	PE	3	--	--	3	30	70	3 Hrs.
DIGITAL IMAGE PROCESSING								
(For ECE)								
Course Objectives: Students are expected to								
1.	Recall and summarize the digital image fundamentals and to be exposed to basic image processing techniques.							
2.	Be familiar with image enhancement, segmentation, compression and morphological processing techniques.							
3.	Illustrate the representation of color images in the form of features.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain digital image fundamentals and basic image processing techniques by analysing image data in spatial and frequency domain.							K3
2.	Evaluate the techniques for image enhancement and restoration.							K3
3.	Define the need for image compression and to analyze various image compression methods.							K3
4.	Experiment the Partition of a digital image into multiple objects using various techniques.							K3
5.	Analyze and manipulate the geometric structure of images and Illustrate the use of different color models to represent an image.							K3
SYLLABUS								
UNIT-I (10Hrs)	Digital Image Fundamentals: Introduction, Origin of Digital Image Processing, Fundamental Steps in Digital Image Processing, Elements of Visual Perception, Image Sensing and Acquisition, Image Sampling and Quantization, Basic Relationships between pixels. Image Transforms: Need for image transforms, 2D Fourier Transform, Discrete Cosine Transform and their properties.							
UNIT-II (10 Hrs)	Image Enhancement and Image Restoration: Intensity Transformations: Basic intensity transformations, Histogram processing – Histogram equalization - Basics of Spatial Filtering: Smoothing and Sharpening of Spatial Filtering. Fundamentals of Image Restoration - Noise models - Mean Filters, Order-Statistic Filters, Adaptive filters: Adaptive local noise reduction filter, Adaptive median filter.							
UNIT-III (10 Hrs)	Image Compression: Fundamentals: Coding Redundancy, Spatial and Temporal Redundancy, Irrelevant Information - Basic Image Compression model - Basic Compression methods: Huffman Coding, JPEG-standard and run length coding.							

UNIT-IV (10 Hrs)	Image Segmentation: Fundamentals - Point, Line and Edge Detection: Detection of isolated points, Line Detection, Basic Edge Detection - Thresholding: Intensity Thresholding, Basic Global Thresholding - Region based Segmentation: Region Growing, Region Splitting and Merging.
UNIT-V (10 Hrs)	Color Image Processing and Morphological Image Processing: Color fundamentals - Color Models: RGB color model, CMY and CMYK color models, HSI color model, Color Complements, Preliminaries of Morphological processing: Opening, Closing, Erosion, Dilation.
Textbooks:	
1.	Rafael C. Gonzales, Richard E. Woods, "Digital Image Processing", Third Edition, Pearson Education, 2010
2.	Digital Image Processing by S Jayaraman, S Esakkirajan, T Veerakumar, Tata McGraw-Hill
Reference Books:	
1.	Anil Jain K. "Fundamentals of Digital Image Processing", PHI Learning Pvt. Ltd., 2011.
2.	Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, "Digital Image Processing Using MATLAB", Third Edition Tata McGraw Hill Pvt. Ltd., 2011.
e-Resources	
1.	https://nptel.ac.in/courses/117/104/117104069/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4111	PE	3	--	--	3	30	70	3 Hrs.
SATELLITE COMMUNICATIONS								
(For ECE)								
Course Objectives:Students are expected to								
1.	Functionality of KEPLAR’ S laws planetary motion.							
2.	Be aware of space segment equipment.							
3.	Understand various parameters of link design.							
4.	Analyze the various multiple access techniques.							
5.	Introduce basic concepts of GPS and Satellite navigation.							
Course Outcomes:At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Identify the necessary components required in modern satellite communication systems.							K3
2.	Design and build space segment, depending upon the requirement.							K4
3.	Design link margin for various applications.							K4
4.	Select the appropriate multiple access technique for effective communication.							K3
5.	Explain the fundamental concepts of GPS and satellite navigation.							K3
SYLLABUS								
UNIT-I (10Hrs)	Orbital Mechanics and Launchers: Orbital Mechanics, Look Angle determination, Orbital perturbations, Orbit determination, launches and launch vehicles, Orbital effects in communication systems performance.							
UNIT-II (10 Hrs)	Satellite Sub-Systems: Attitude and orbit control system, telemetry, tracking, Command and monitoring, power systems, communication subsystems, Satellite antenna Equipment reliability and Space qualification.							
UNIT-III (10 Hrs)	Satellite Link Design: Basic transmission theory, system noise temperature and G/T ratio, Design of down links, up link design, Design of satellite links for specified C/N, System design example.							
UNIT-IV (8 Hrs)	Multiple Access: Frequency division multiple access (FDMA), Time division Multiple Access (TDMA) Frame structure, Examples. Satellite Switched TDMA Onboard processing, DAMA, Code Division Multiple access (CDMA), Spread spectrum transmission and reception.							
UNIT-V	Satellite Navigation & The Global Positioning System: History of GPS, Evolution of							

(12 Hrs)	GPS, Development of NAVSTAR GPS, GPS working principle, GPS Configuration, Satellite Signal Generation, Signal Power, and Other Global Navigation Systems.
Textbooks:	
1.	Satellite Communication, by Timothy Pratt, Charles Bostian, Jeremy Allnutt(Second Edition), John Wiley & Sons.
2.	Global Navigation Satellite systems by G.S.RAO, McGraw Hill.
Reference Books:	
1.	Satellite Communications, by Dennis Roddy (Fourth edition), McGraw Hill.
2.	Satellite Communication Systems Engineering, by Wilbur L. Pritchard, Henri G. Suyderhoud, Robert A. Nelson (Second Edition), Pearson.
e-Resources	
1.	https://www.youtube.com/results?search_query=Orbital+Mechanics+and+Launchers)
2.	https://www.youtube.com/watch?v=lvKOe4hivJU



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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC4112	PE	3	--	--	3	30	70	3 Hrs.
LOW POWER VLSI DESIGN								
(For ECE)								
Course Objectives: Students are expected to								
1.	Provide a basic idea on different low power circuit design techniques							
2.	Identify the power dissipation mechanisms in various MOS logic styles							
3.	Familiarize suitable techniques to reduce power dissipation							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the sources of power dissipation in digital IC systems							K2
2.	Understand the impact of power on system performance and reliability							K2
3.	Understand leakage sources and reduction techniques							K2
4.	Recognise and acquaint with the advanced issues in VLSI systems, specific to the deep-submicron silicon technologies							K2
5.	Acquaint with the mechanisms of power dissipation in CMOS integrated circuits							K2
SYLLABUS								
UNIT-I (10Hrs)	Physics of Power dissipation in MOSFET devices MIS structure, Need for low power circuit design, Threshold voltage, body effects, Short channel effects-surface scattering, punch through, velocity saturation, impact ionization, Hot electron effects, Drain induced barrier lowering, narrow width effects.							
UNIT-II (10 Hrs)	Sources of power dissipation in CMOS-Switching power dissipation: Short circuit power dissipation, glitching power dissipation, Leakage power dissipation, Transistor leakage mechanisms of deep submicron transistors.							
UNIT-III (10 Hrs)	Circuit techniques for leakage power reduction – standby leakage control using transistor stacks, multiple Vth techniques, Dynamic Vth techniques, supply voltage scaling techniques, Deep submicron devices design issues, Minimizing short channel effect.							
UNIT-IV (8 Hrs)	Design and test of low voltage CMOS – Circuit design style, clocked design style- Basic concept, Domino logic (domino NAND gate), Differential Current Switch Logic.							
UNIT-V (12 Hrs)	Non clocked circuit design style-fully complementary logic, NMOS and pseudo –NMOS logic, differential cascade voltage switch logic (DCVS), Pass transistor logic.							

Textbooks:	
1.	KiatSeng Yeo, Samir S. Rofail, Wang-Ling Goh, “CMOS/Bi CMOS ULSI Low Voltage Low Power”, Pearson Education Asia 1st Indian reprint, 2002
2.	Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley India,2000.
Reference Books:	
1.	Rabaey, Pedram, “Low Power Design Methodologies” Kluwer Academic
2.	Yeo, “CMOS/BiCMOS ULSI Low Voltage Low Power” Pearson Education 2. GrayYeap, Practical low power digital VLSI design, Springer, 1998
3.	GrayYeap, Practical low power digital VLSI design, Springer, 1998
e-Resources	
1.	https://archive.nptel.ac.in/courses/106/105/106105034/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
	SEC	--	--	3	1.5	30	70	3 Hrs.
DIGITAL SIGNAL & IMAGE PROCESSING LAB								
(for ECE)								
Course Objectives:								
1	To implement Convolution and Transform techniques on a given sequence							
2	To analyze frequency response of Discrete Time Signals and Systems							
3	To design different Digital filters							
4	To implement basic Image Processing techniques							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Make use of MATLAB simulation tool for performing various operations on discrete signals.							K4
2	Make use of MATLAB simulation tool to verify different DSP algorithms.							K4
3	Make use of MATLAB simulation tool to perform various operations on an Image.							K4
SYLLABUS								
1	Verification of Sampling Theorem							
2	Generation of DiscreteTime (DT) Sequences and Signals							
3	Linear Convolution & Circular Convolution							
4	Frequency Domain Analysis of DT Signals & Systems using DTFT							
5	LTI System Simulation							
6	Design and Verification of IIR Digital Filters							
7	Design and Verification of FIR Digital Filters using Windows							
8	Spectrum Analysis using Fast Fourier Transform(FFT)							
9	Multirate Up sampling & Down sampling							
10	Design and Simulation of Interpolation & Decimation Filters							
11	Generation & Detection of DTMF Signals							
12	Image intensity transformations							
13	Histogram processing							
14	Filtering images in spatial and frequency domain							
15	Computation of N-point DFT of a given sequence using DSP processor							
16	Computation of FFT of a given sequence using DSP processor							
Reference Books:								
1	DS Pusing MATLAB by JohnG. Proakis.							

2	Digital Image Processing by Gonzalez and Woods
3	Sanjit K. Mitra, "Digital Signal Processing: A Computer Based Approach", TataMcGrawHill.
4	Lab manual
e-Resources:	
1	https://www.mathworks.com/academia/books/digital-image-processing-gonzalez.html
2	http://vlabs.iitkgp.ernet.in/dsp/index.html



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Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	--	--
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives:Students are expected								
1.	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.							
2.	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.							
3.	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.							K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.							K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.							K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grassroots governance.							K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.							K3
SYLLABUS								
UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (6Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							
UNIT-III	State Government and its Administration Governor - Role and Position - CM and							

(6Hrs)	Council of ministers, State Secretariat: Organisation, Structure and Functions.
UNIT-IV (6Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass root democracy.
UNIT-V (6Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



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Regulation: R23		IV / IV - B.Tech. II - Semester							
ELECTRONICS & COMMUNICATION ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23EC4201	Project and Internship	PR	--	--	24	12	60	140	200
TOTAL			--	--	24	12	60	140	200



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Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23EC4201	PR	--	--	24	12	60	140	3 Hrs.
PROJECTWORK								
(For ECE)								
Course Objectives:								
1	To provide an opportunity to work in group on a topic/ problem / experimentation							
2	To encourage creative thinking process							
3	To provide an opportunity to analyze and discuss the results to draw conclusions							
4	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions, and decision-making process.							
Course Outcomes: At the end of the course the students will be able to								
S.No.	Outcome							Knowledge Level
1	Identify a current problem through literature/field/case studies							K3
2	Identify the objective and methodology for solving the problem							K3
3	Design and Develop technology/process for solving the problem							K4
4	Evaluate the technology/process							K5
*The object of Project Work is to enable the student to take up investigative study in the broad field of Electronics and Communication Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work.								
The assignment to normally include:								
a) Survey and study of published literature on the assigned topic.								
b) Working out a preliminary approach to the problem relating to the assigned topic.								
c) Conducting preliminary Analysis/Modeling/Simulation/Experiment/Design/ Feasibility.								
d) Preparing a written report on the study conducted for presentation to the department.								
e) Final Seminar, as oral Presentation before a departmental committee.								